Figure 30.—Point counter (*after Geiger*).

Under certain conditions, when the voltage is not too high, the multiplication of the number of electrons always increases by the same factor. Therefore we speak of a *proportional region* of the counter and also of a *proportional count.* The factor just mentioned may be of the order of magnitude of 1,000 or even higher. But if the voltage on the counter is increased beyond a certain limit, we pass beyond the proportional region. In that case, the electrons liberated by the particle start a genuine glow discharge, so that the result is a ten-millionfold or even a hundred-millionfold increase. In that case the discharge must be stopped, so that the counter may once again be ready for a new particle. In this *resolving region*, the amplification is independent of the number of the primarily liberated electrons. The amplification continues always until the very moment when a glow discharge begins to take place.

About fifteen years ago, this point counter was considerably improved by Geiger and Müller, and has become the counter which is still by far the most important observing apparatus of the nuclear physicist. In principle it is very similar to the original Geiger point counter, except that instead of a point, a thin wire is placed in its centre. (Figure 31.) Usually, it is filled not with air, but with a mixture of argon, under a pressure of 60 to 80 mm. of mercury, and alcohol vapour under a pressure of about 10 mm. of mercury. But there are also several variants. The wire is earthed through a very big resistance, and the outer cover has a potential difference of 1,000–1,200 volts relative to the earth.

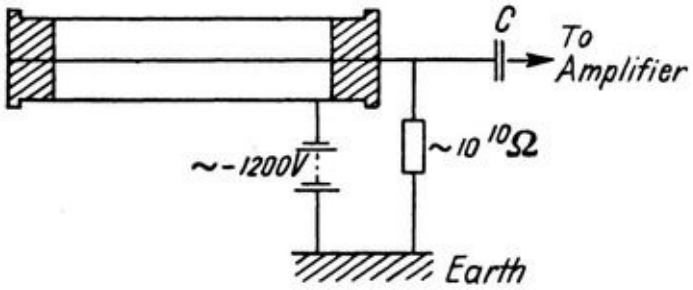



Figure 31.—Geiger-Müller counter.

The situation here is similar to that in the point counter. At a lesser voltage, a proportional amplification, by the factor 1,000, takes place. When the voltage is higher, a glow discharge begins; the device is now operating in the resolving region. From the moment of the inception of the glow discharge, the wire, which may be connected to a condenser, becomes strongly charged, as does also the condenser, since owing to the very high resistance, an appreciable time must elapse before the charge flows to the earth. Therefore,